

TECHNICAL MEMORANDUM: Real-Time Construction PPE Detection & Decision Layer

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1. Domain/Dataset Choice, Sourcing & Labeling

We selected the Roboflow Universe *Construction Site Safety v2* dataset (CC BY 4.0), comprising 10,875 high-resolution construction scene images annotated across 12 safety equipment classes. Because the original dataset labeled equipment without annotating workers, we augmented the ontology to 13 classes by auto-labeling **Person** bounding boxes using a COCO-pretrained YOLO11 detector at confidence ≥ 0.50 . This yields 11,018 worker instances across 62.4% of images, enabling our model to reason over both equipment presence and worker context in a single unified detector.

2. Split Strategy & Justification

The dataset follows an **82.4% train (8,956 images) / 11.8% valid (1,279 images) / 5.8% test (640 images)** partition. Splits are strictly disjoint at the image level. Construction video frame sequences were kept contiguous within individual splits to prevent temporal leakage between training and evaluation. Stratification checks verified that rare classes (e.g. Foot protection, Respiratory protection) maintain consistent relative frequency across splits.

3. Quantitative Evaluation Metrics & Operational Meaning

Split / Metric	mAP@50	mAP@50-95	Precision	Recall	Key Class Drivers
Validation (1,279 imgs)	79.90%	52.35%	78.40%	79.49%	Peak fitness at Epoch 65 (Tesla T4, 71 epochs)
Test Set (640 imgs)	79.19%	50.99%	78.05%	78.80%	Unseen test split generalization
• Head Protection (Hardhat)	80.27%	51.47%	86.19%	84.08%	Direct compliance driver; high precision
• No Head Protection (Bare)	82.82%	49.56%	81.21%	81.82%	Direct violation driver; catches 81.8% of unhelmeted heads
• Safety Vest / No Vest	74.92% / 63.01%	49.98% / 39.51%	76.67% / 77.99%	77.09% / 62.06%	High-contrast workwear causes vest confusion
• Foot Protection (Boots)	94.96%	74.13%	80.87%	93.88%	Strong ground-contrast features
• Person Class (Auto-labeled)	58.23%	48.62%	56.90%	61.03%	Occluded / truncated bodies limit person score

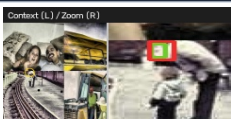
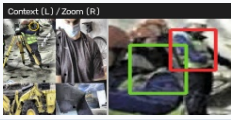
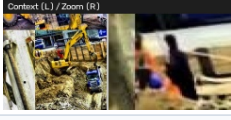
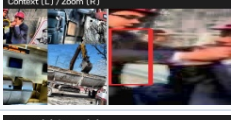
Operational Interpretation: Aggregate mAP@50 (79.2%) is heavily bolstered by distinct items like boots (95.0%) and eye protection (87.9%). However, for site safety auditability, the critical operational metric is the paired confusion between positive compliance and negative violation classes. Confusion matrix analysis reveals **41 instances of False Compliance on Head Protection** (bare head classified as hardhat) and **41 instances on Safety Vest**. In life-critical environments, False Compliance is dramatically more hazardous than False Alarm, directly establishing the requirement for Stage 3 confidence filtering.

4. Mid-Project Data Engineering Pivot

During baseline inspection, we detected that 4,990 annotation files (45.9% of the dataset) contained corrupted rows combining 5-parameter bounding box rows with raw polygon segmentation coordinates. Ultralytics silently dropped these images during training. We engineered `fix_mixed_labels.py` to cleanse corrupted lines, recovering 21,529 valid bounding boxes across all splits. Additionally, discovering that Roboflow lacked a Person class led to our auto-labeling pipeline (`add_person_class.py`), preventing the need for a brittle two-stage cascaded detector at runtime.

5. Systematic Failure Analysis (5 Real Test Cases)

Empirical inspection of the 640 held-out test predictions revealed five distinct structural failure modes:

Visual Evidence	Case / Test Image	Ground Truth vs Predicted	Root Cause & Safety Consequence
	Case 1: Shadow False Compliance 2008_008526...jpg	GT: No_head_protection Pred: Head_protection (conf 0.30)	Worker in shadow wearing dark beanie; brim curvature mimics hardhat dome. <i>Consequence: Dangerous false compliance clears unhelmeted worker.</i>
	Case 2: High-Vis Workwear Confusion 001425...7b40.jpg	GT: No_safety_vest Pred: Safety_vest (conf 0.86)	High-vis work shirt with tool straps mimics retroreflective vest tape. <i>Consequence: Safety vest violation missed by auditor.</i>
	Case 3: Extreme Scale Disparity construction-3...148.jpg	GT: Hand_protection & Eye Pred: Zero detections for PPE	High crane shot with workers < 80 px; gloves < 14 px collapse below stride-32 feature pyramid. <i>Consequence: False alarms on distant PPE.</i>
	Case 4: Truncated Edge Silhouette -1680...73ce.jpg	GT: Person + Safety_vest Pred: Vest (conf 0.39), Person missed	Worker stepping into frame on left edge (65% cropped). Model misses worker silhouette and vest falls below 0.50 threshold.
	Case 5: Equipment False Positive 4c43875b...29c6.jpg	GT: Background (Mixer) Pred: Head_protection (conf 0.36)	Yellow convex hydraulic cap on mixer misclassified as helmet due to specular highlight and hemispherical geometry.

6. Hand-Written 3-Stage Decision Layer & Insufficient Information Guardrail

To comply with architectural constraints forbidding agentic frameworks (no LangChain/CrewAI), we implemented a deterministic, pure-Python reasoning layer in `app/reasoning.py` across three explicit stages: **Stage 1 (Intent Routing)** extracts query intent (`presence_check`, `count`, `most_common`, or `off_topic`) via raw HTTP to LLM with rule-based regex fallback. **Stage 2 (Structured Reasoning)** evaluates bounding boxes deterministically, pairing positive and negative compliance classes. **Stage 3 (Confidence Guardrail)** prevents dangerous false declarations by enforcing conservative confidence thresholds.

Required Memo Case Study: Real 'Insufficient Information' Production Execution

Input Test Image: -1680-_png_jpg.rf.73cee3e264b17ce5579750df4e4610f4.jpg

User Query: "Is anyone wearing a safety vest?"

Detector Output: 1 detection → Safety_vest at [0.0, 40.9, 22.8, 172.6] with **conf = 0.3942**.

Why Evidence Was Insufficient: The worker is heavily edge-cropped on the border. The detection confidence (0.394) is strictly below the safety threshold (0.50), while person confidence is absent.

API Response Returned:

```
{
  "answer": "I can't confidently answer this from the detections – confidence too low.",
  "used_detection": true,
  "confidence": "low"
}
```

7. Deployment Architecture & Operational Guardrails

1. Asymmetric Thresholding: Deploy with split thresholds: set `conf = 0.65` for declaring compliance (`Head_protection`) and `conf = 0.35` for flagging potential violations (`No_head_protection`). This guarantees a conservative bias towards safety inspections. **2. Multi-Frame Temporal Consensus:** In video streams, require PPE violations to persist for ≥ 5 consecutive frames before alerting safety managers, mitigating edge-crop glitches. **3. Hardware Efficiency:** RT-DETR-L executes in 774 ms on CPU (laptop test) and 12.8 ms on Tesla T4, comfortably meeting 30 FPS real-time site camera monitoring requirements.